TESS Light-Curves Workflow

Juan David Galan Vargas

Astro-Informatics 1
Magister en Astronomía
Universidad de Antofagasta

20 / 07 / 2026

What is TESS?

- **Mission:** Survey 500.000 of G, K, and M type Main Sequence stars (brighter than 12 magnitudes) to detect **exoplanets** from Earth-size to gas giants.
- **Scope:** 85% of the sky in a originally 2 year planned mission, which has been extended for over 8 years.
- **Transit Method:** Detection of periodic dips in the luminosity of a star caused by an orbiting planet blocking their host star's light [3].

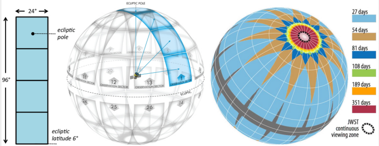

Figure: Division of the sky in 26 observation sectors. The spacecraft will spends two 13.70-day orbits observing each sector each time. Taken from: `https://tess.gsfc.nasa.gov`

TESS Instruments

- Package of four wide-field-of-view charge-coupled device (CCD) cameras.
  - Each camera features four low-noise, low-power 4 megapixel CCDs.
  - The four CCDs are arranged in a 2 x 2 detector array for a total of 16 megapixels.
- Cameras have a bandpass range of 600 nm (orange) to 1000 (NIR) nm [5].
- Every 13.70 days TESS transmits the data it collected during its most recent orbit back to earth [4].
  - This transmission occurs over a period of approximately three

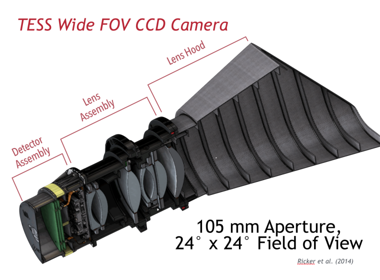

Figure: TESS camera diagram.

Data

Downloading Shell Script

How to obtain the Data?

We used the Mikulski Archive for Space Telescopes (MAST) to obtain a Shell Script to download the TESS data easily.

How to obtain the Data?

We used the Mikulski Archive for Space Telescopes (MAST) to obtain a Shell Script to download the TESS data easily.

tesscurl sector 73 lc.sh

The Shell Script downloads the light curve data of the sector 73 (which correspond to the third survey of the sector 21). The downloaded data is provided in FITS file format.

Downloading Shell Script

How to obtain the Data?

We used the Mikulski Archive for Space Telescopes (MAST) to obtain a Shell Script to download the TESS data easily.

tesscurl sector 73 lc.sh

The Shell Script downloads the light curve data of the sector 73 (which correspond to the third survey of the sector 21). The downloaded data is provided in FITS file format.

TOPCAT

The downloaded FITS file then were saved as CSV files using TOPCAT. It should be noted that this must done file by file with this method.

Files Name List

```
output-file="csv-files-names.txt"
ls *.csv > "$output-file"
```

Batch Name List

```
input-file="csv-files-names.txt"
num-names=5

split -l $num-names -d –additional-suffix=.txt "$input-file" parte-
ls parte-*.txt
```

Format Conversion

Change of Extension: csv –> lc

```
SOURCE-DIR= [insert here path]
for file in "$SOURCE-DIR"/*.csv; do
[ -f "$file" ] || continue
base=$(basename "$file" .csv)
output="./$base.lc" #Change of extension
```

Change of Extension: csv –> lc

```
SOURCE-DIR= [insert here path]
for file in "$SOURCE-DIR"/*.csv; do
[ -f "$file" ] || continue
base=$(basename "$file" .csv)
output="./$base.lc" #Change of extension
```

Change of Delimiter: , –> " "

```
awk -F ',' '
```

Keep Relevant Columns

```
NR==1 {for (i=1; i<=NF; i++) { gsub(/^[ t"]+|[ t"]+$/, "", $i)
if ($i=="TIME") c1=i
if ($i=="TIMECORR") c2=i
if ($i=="SAP-FLUX") c3=i
if ($i=="SAP-FLUX-ERR") c4=i
if ($i=="PDCSAP-FLUX") c5=i
if ($i=="PDCSAP-FLUX-ERR") c6=i}
print "TIME TIMECORR SAP-FLUX SAP-FLUX-ERR PDCSAP-FLUX
PDCSAP-FLUX-ERR"
next}
{print $c1, $c2, $c3, $c4, $c5, $c6}
' "$file" > "$output"
echo "Created $output"
done
```

Distinguishing Signal from Instrument Noise

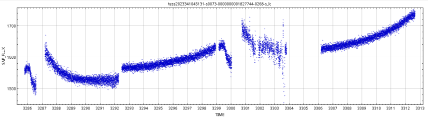

Figure: SAP-FLUX Light Curve: Raw flux summing calibrated pixels. Reveals massive instrumental artifacts that mask any signal.

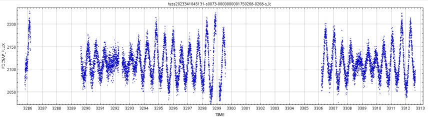

Figure: PDCSAP-FLUX Light Curve: Preresearch data conditioned. Correction of instrumental noise reveals a periodic signal [4].

Distinguishing Signal from Instrument Noise

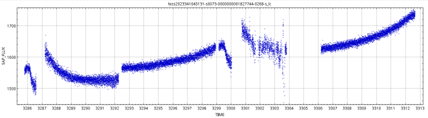

Figure: SAP-FLUX Light Curve: Raw flux summing calibrated pixels. Reveals massive instrumental artifacts that mask any signal.

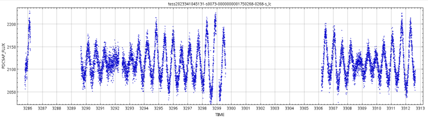

Figure: PDCSAP-FLUX Light Curve: Preresearch data conditioned. Correction of instrumental noise reveals a periodic signal [4].

The following workflow steps will use PDCSAP-FLUX Light Curves.

Light Curve Analysis

Outlier Criteria

The criteria selected to consider a data point an outlier was that their PDCSAP-FLUX value was 3 σ away from the median value.

Outlier Criteria

The criteria selected to consider a data point an outlier was that their PDCSAP-FLUX value was 3 σ away from the median value.

A More Robust σ

The Median Absolute Deviation, which is defined as the median of the absolute deviations from the data's median. This estimator can be approximate to σ by multiplying by scaling factor (which assuming the data has a normal distribution it has value of **1.4826** [2].)

Lomb-Scargle Periodogram

- Detect periodic signals in **unevenly spaced observations**.
- Least-squares fit of sinusoidal functions across various frequencies [6].
- Not the most optimal method to identify transit frequencies.
  - Planetary transit is a short, sharp "dip" in a light curve.
  - Other methods that fit this shape exits such as BLS or even TLS which takes into account the **Limb Darkening** effect [1].

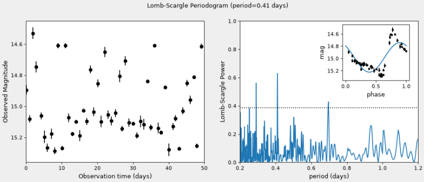

Figure: Lomb-Scargle application on a 50 nightly observations of a simulated RR Lyrae-like variable star. Taken from:

Light Curve Analysis

Visualization: Phase Space Light Curves and Periodogram.

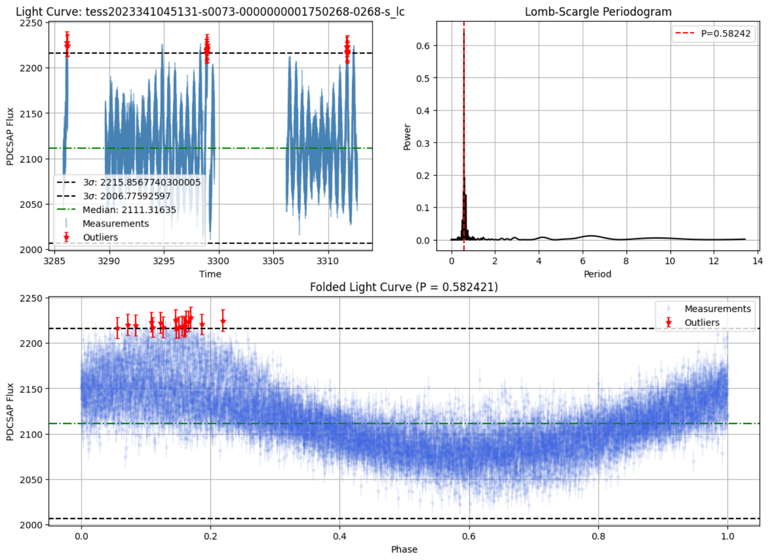

Figure: Light Curve with sinusoidal shape.

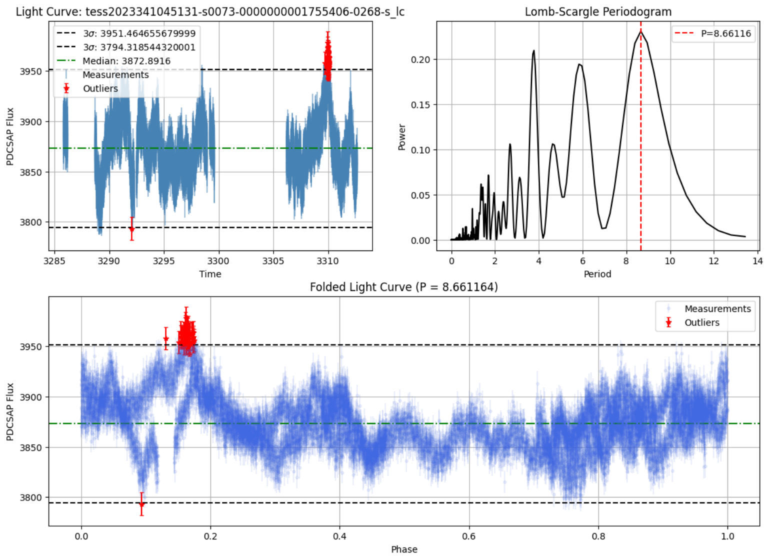

Figure: Light Curve with possible multiple sources.

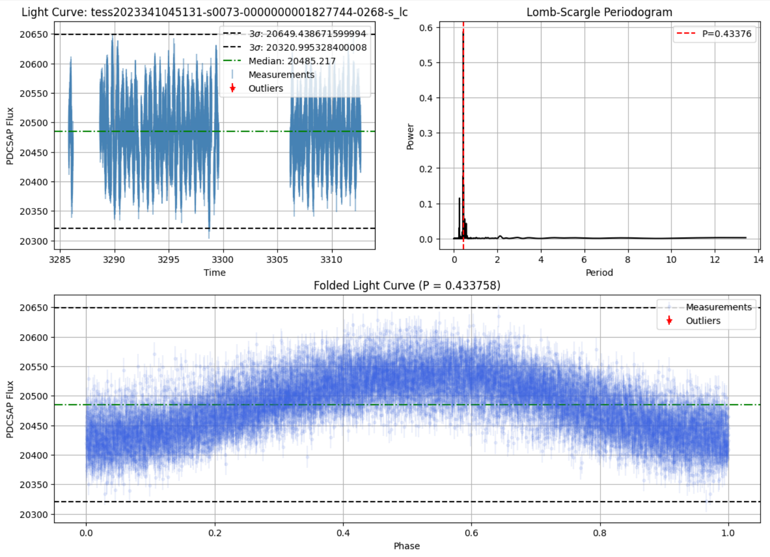

Figure: Light Curve with sinusoidal shape.

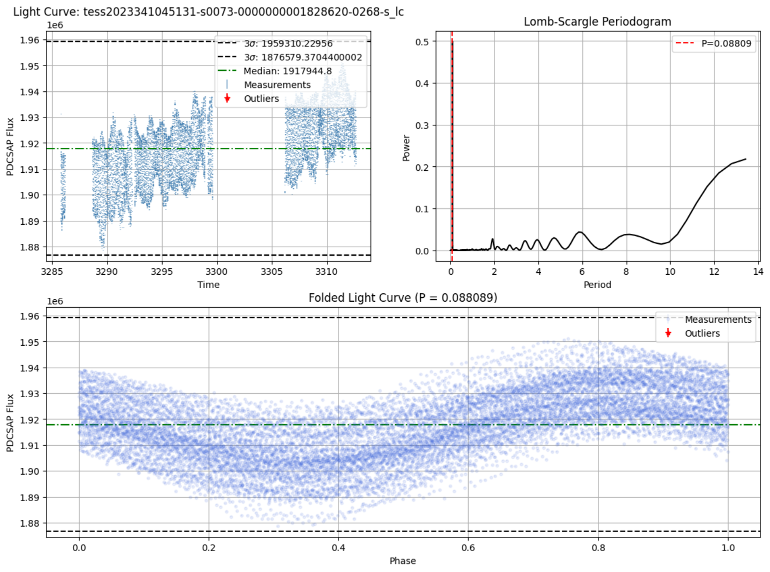

Figure: Light Curve with weird shape.

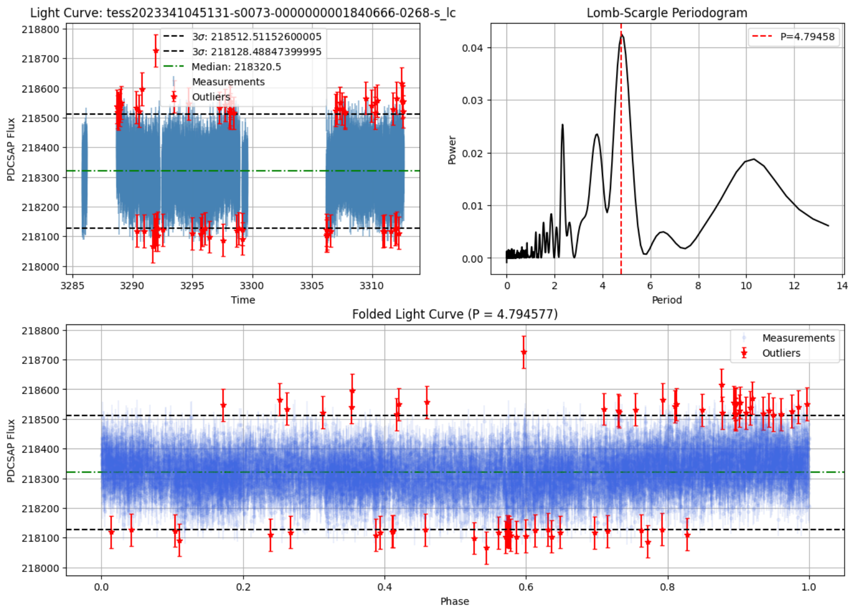

Figure: Light Curve with no signal.

Conclusions

TESS data (along many other datasets thanks to the MAST) is easily available for scientific use. It only requires basic use of terminal commands (Shell Scripts to download.)

TESS data (along many other datasets thanks to the MAST) is easily available for scientific use. It only requires basic use of terminal commands (Shell Scripts to download.)

Converting FITS files into CSV format via TOPCAT is not efficiente for a large dataset. This can be done with Astropy.Table:

for fits-path in fits-files:
table = **astropy.Table.read**(fits-path, hdu=1)
base-name = **os.path.splitext**(**os.path.basename**(fits-path))[0]
csv-path = **os.path.join**(directory, f"base-name.csv")
astropy.table.write(csv-path, format='ascii.csv', overwrite=True)

TESS data (along many other datasets thanks to the MAST) is easily available for scientific use. It only requires basic use of terminal commands (Shell Scripts to download.)

Converting FITS files into CSV format via TOPCAT is not efficiente for a large dataset. This can be done with Astropy.Table:

for fits-path in fits-files:
table = **astropy.Table.read**(fits-path, hdu=1)
base-name = **os.path.splitext**(**os.path.basename**(fits-path))[0]
csv-path = **os.path.join**(directory, f"base-name.csv")
astropy.table.write(csv-path, format='ascii.csv', overwrite=True)

Thoughtful visual design of the plots (both radiograms and light curves) improves significantly its readability.

Thanks for your attention!

References I

Hippke, Michael and Heller, René.
Optimized transit detection algorithm to search for periodic transits of small planets.
AA, 623:A39, 2019.

Christophe Leys, Christophe Ley, Olivier Klein, Philippe Bernard, and Laurent Licata.
Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median.
Journal of Experimental Social Psychology, 49(4):764–766, July 2013.

National Aeronautics and Space Administration.
TESS: Transiting Exoplanet Survey Satellite.
Fact Sheet FS-2014-1-120-GSFC, NASA Goddard Space Flight Center, October 2014.
Archived from the original PDF on December 17, 2014.

National Aeronautics and Space Administration.
TESS Observatory Guide.
NASA, 2018.
PDF document. Public domain.

George R. Ricker, Joshua N. Winn, Roland Vanderspek, David W. Latham, Gáspár Á. Bakos, Jacob L. Bean, Zachory K. Berta-Thompson, Timothy M. Brown, Lars Buchhave, Nathaniel R. Butler, et al.
Transiting exoplanet survey satellite.
Journal of Astronomical Telescopes, Instruments, and Systems, 1(1):014003, January 2015.

Jeffrey D. Scargle.
Studies in astronomical time series analysis. II. Statistical aspects of spectral analysis of unevenly spaced data.
The Astrophysical Journal, 263:835–853, 1982.